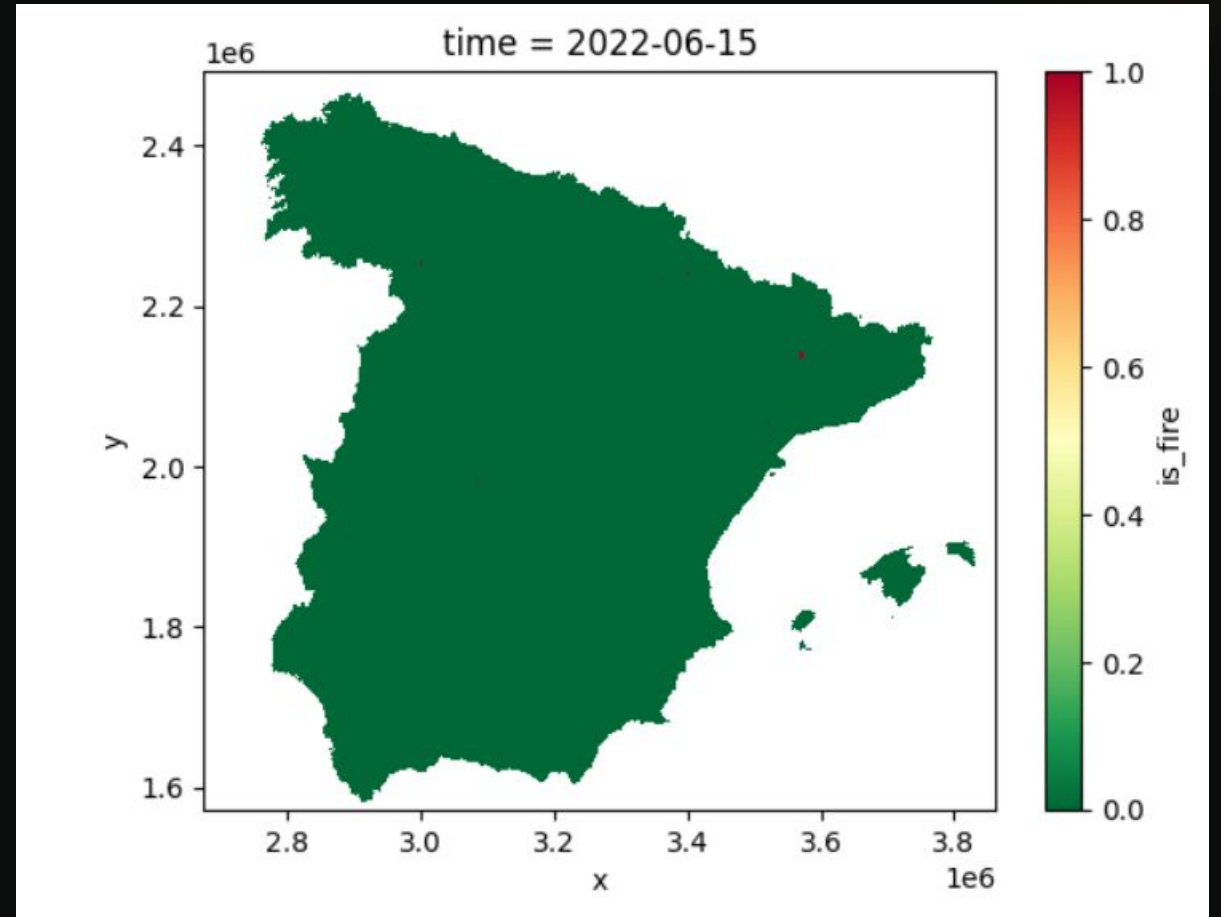


Deep Learning for wildfire risk prediction in Spain

Shuangjie Xia, Felipe López, Melker Svensson

State of the Art

- **IberFire Spatio-Temporal Cube:** Introduced in 2025 by Ercibengoa et al. High-resolution 1km $\times$ 1km daily datacube covering mainland Spain from 2007 to 2024.
- **260 Features:** Across 8 classes including weather, geography, fire history, human activity, and land cover.
- **The Baseline Issue:** reported 0.955 ROC-AUC but validated on artificially balanced subsets, obscuring real-world imbalance and performance on false negatives.



IberFire dataset

- Coordinates: x, y, time
- Target: is_fire
- Fire Weather Index (FWI)
- Features: geographic, meteorological, vegetation,...

► Dimensions: (y: 920, x: 1188, time: 6241)

▼ Coordinates:

y	(y)	float64	2.492e+06 2.491e+06 ... 1.573e+06	 
x	(x)	float64	2.675e+06 2.676e+06 ... 3.862e+06	 
time	(time)	datetime64[ns]	2007-12-01 ... 2024-12-31	 

▼ Data variables:

x_index	(y, x)	uint16	...	 
y_index	(y, x)	uint16	...	 
is_spain	(y, x)	uint16	...	 
is_fire	(time, y, x)	uint8	...	 
is_near_fire	(time, y, x)	uint8	...	 
x_coordinate	(y, x)	float32	...	 
y_coordinate	(y, x)	float32	...	 
is_sea	(y, x)	uint8	...	 
is_waterbody	(y, x)	int32	...	 
AutonomousCo...	(y, x)	uint16	...	 

| Data sampling from IberFire

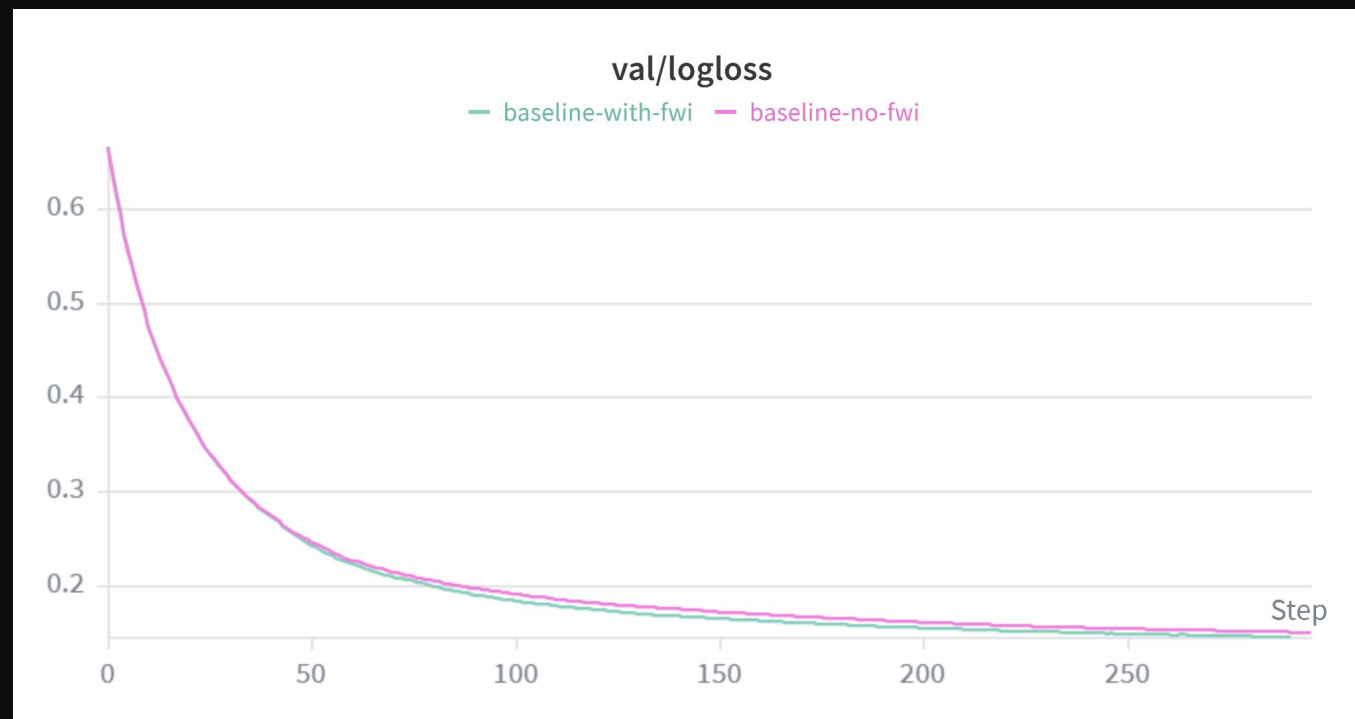
- 0.002% fire instances!
- Negative sampling for training (1:3 ratio)
 - Random negatives
 - Spatial hard negatives
 - Temporal hard negatives
- Negative sampling for val-test (1:10 ratio)
 - Random negatives
- Chronological splits: (2008-2022), (2023), (2024)
- Cleaning...

| Model architectures

- **XGBoost baseline** with binary logistic loss and "scale_pos_weight".
 - baseline with/without fire weather index (FWI)
- **LSTM** with multiple configurations (hidden size, number of layers, learning rate) and BCE loss.
 - Focal loss variant
 - Dropout variant
- **Best LSTM** with/without FWI
- **Ensemble** (XGBoost + LSTM)

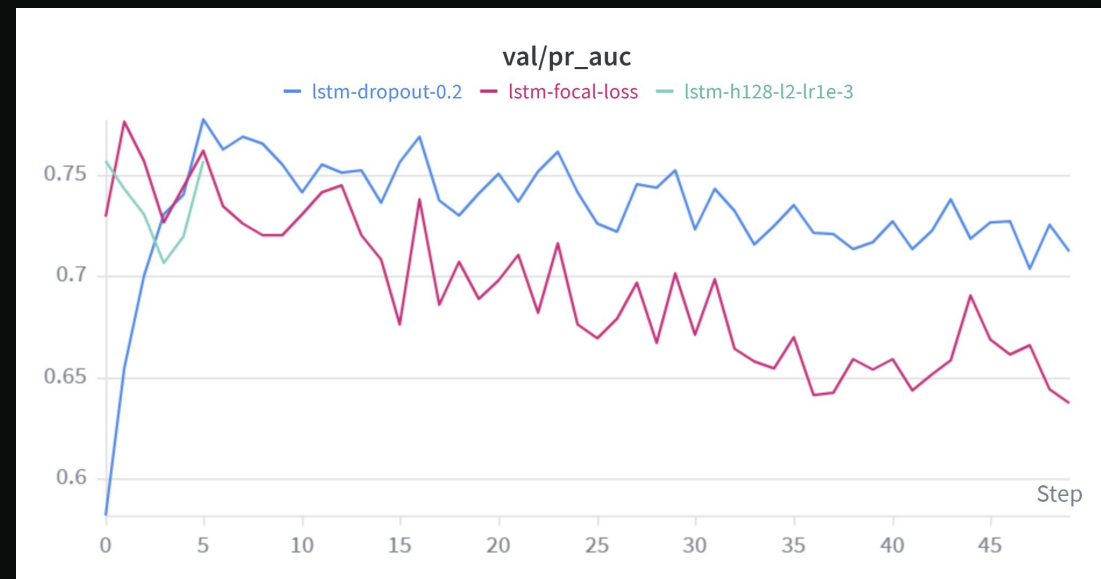
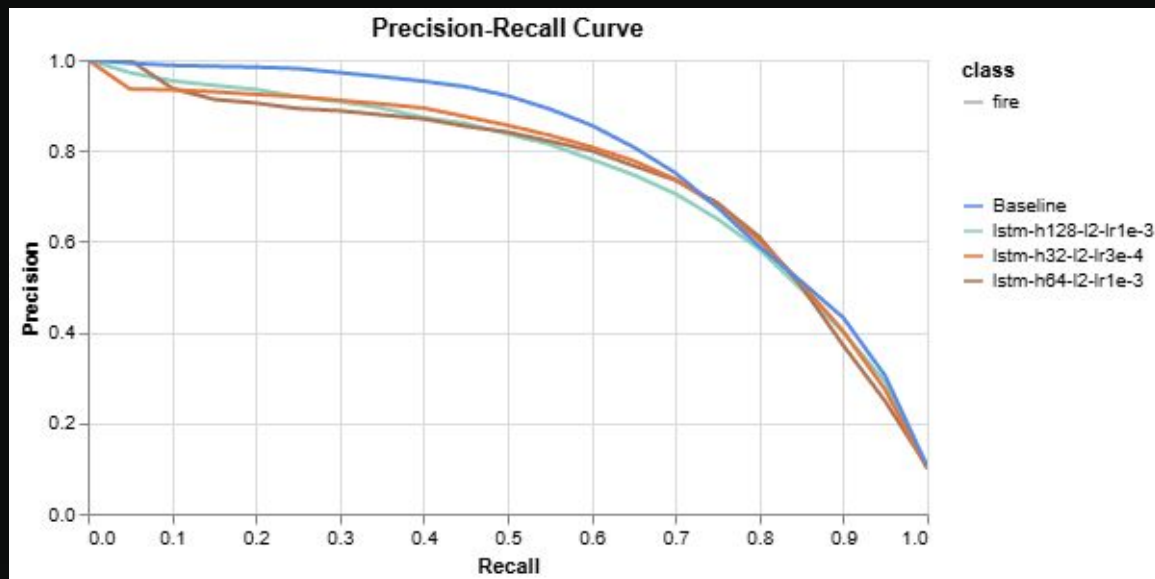
Prioritizing PR-AUC!

XGBoost



Model	PR-AUC	ROC-AUC	Recall	Precision	F2	FN ↓	FP
baseline_no_fwi	0.778	0.956	0.845	0.494	0.740	1,071	5,977
baseline_with_fwi	0.805	0.960	0.853	0.509	0.751	1,012	5,689

LSTMs

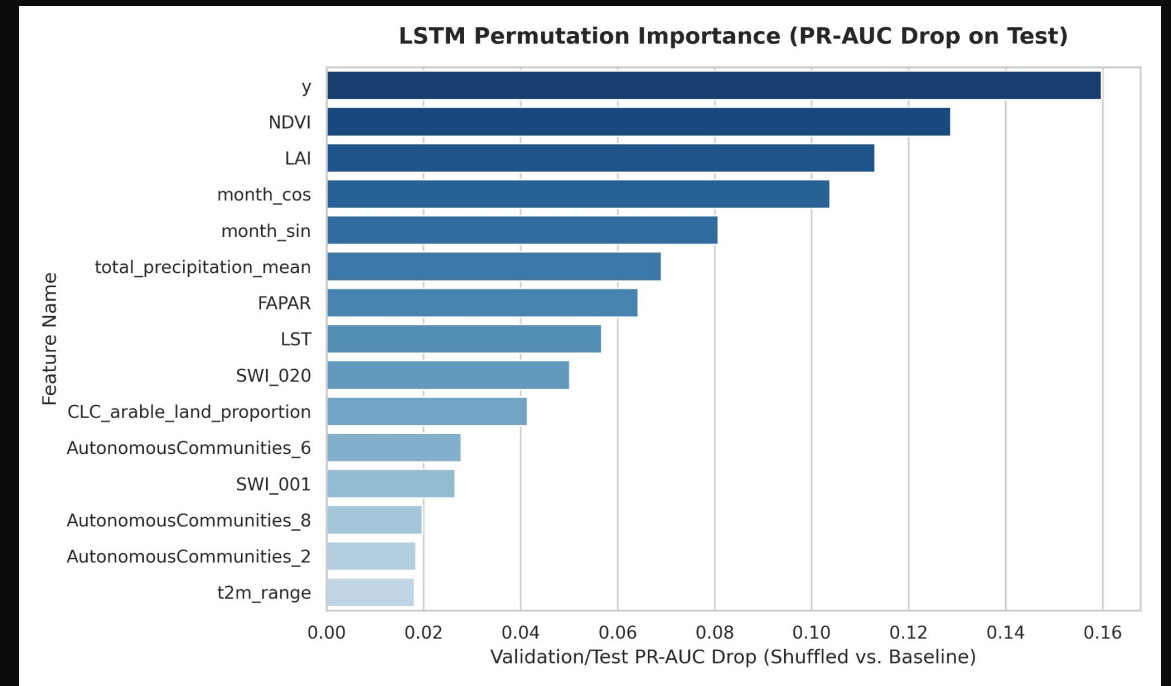
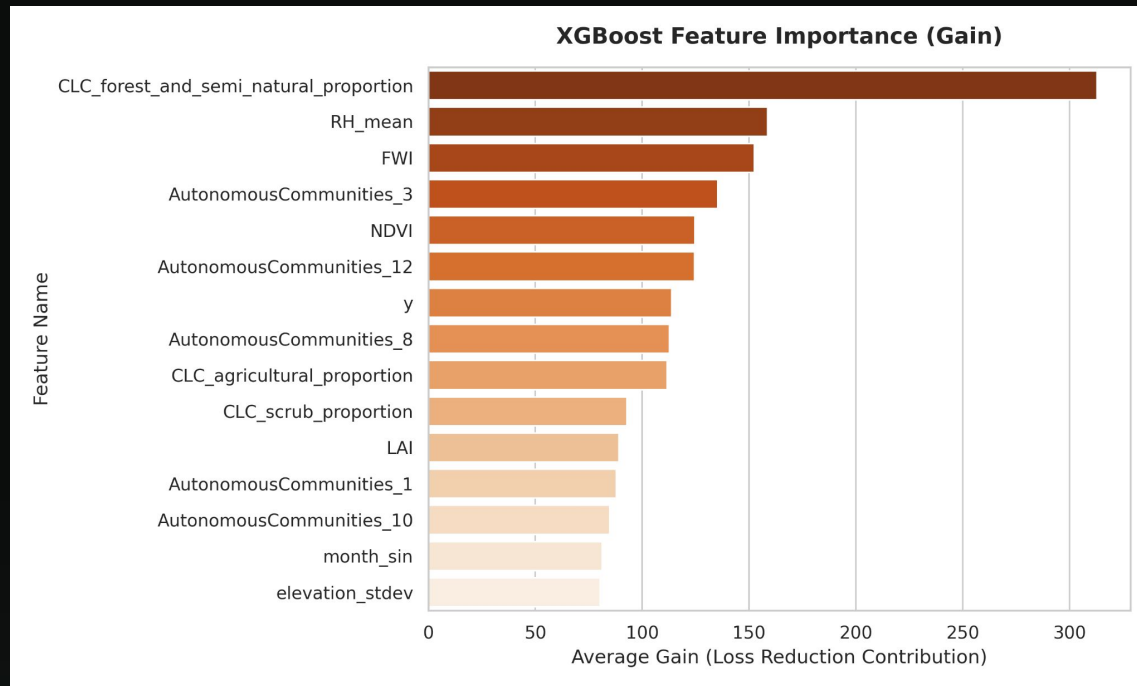


Model	PR-AUC	ROC-AUC	Recall	Precision	F2	FN ↓	FP
best_ablation_no_fwi	0.787	0.958	0.826	0.585	0.763	1,202	4,034
best_ablation_with_fwi	0.774	0.956	0.839	0.573	0.768	1,109	4,315

Comparison table

Model	PR-AUC	ROC-AUC	Recall	Precision	F2	FN ↓	FP
XGBoost	0.805	0.960	0.853	0.509	0.751	1,012	5,689
LSTM ablation (h128, l2, lr1e-3)	0.757	0.954	0.820	0.553	0.748	1,244	4,568
LSTM focal ($\alpha=0.75$, $\gamma=2.0$)	0.776	0.954	0.819	0.565	0.752	1,247	4,356
LSTM dropout (p=0.2)	0.777	0.957	0.824	0.576	0.759	1,213	4,189
LSTM best ablation (no FWI)	0.787	0.958	0.826	0.585	0.763	1,202	4,034
Ensemble (XGB + LSTM focal)	0.845	0.967	0.850	0.606	0.786	1,038	3,816

Discussion: why was XGBoost better than LSTMs?



| Randomness and binary classification

- Wildfires require **ignition**, which is largely random.
- Our models penalize "fire" predictions in high-risk conditions
 - Some actual "no fire" could be near "fire"



| Conclusions

- A more realistic evaluation of IberFire was performed
- Combination of LSTM and XGBoost was the best configuration, achieving 84% PR-AUC.

| Future work

- Increase the time range in LSTMs
- Reframe binary classification
- Try different architectures

Thanks for your attention!

Any questions?